NIIT ENTERPRISE

New Relic APM – 5-Day Advanced Training Plan

Production Support Track

> **Audience:** Production Support teams already familiar with New Relic fundamentals, agent setup, and general administration.
> **Focus:** Transactions and tracing, performance optimization and dashboards, synthetic monitoring (create, manage, troubleshoot), advanced NRQL, integrations, and a capstone.
> **Format:** Instructor-led with scenario-based hands-on labs in every module.

Day 1: Deep Dive into Transactions, Traces, and Services

Module 1: Transaction Analysis for Production Troubleshooting

- Web vs non-web (background) transactions: reading the APM Summary during an incident
- Transaction traces: identifying bottlenecks, slow segments, and N+1 patterns
- Segments, call trees, and breakdown tables – where time is really spent
- External services and database calls: latency, throughput, and failure impact
- Key transactions and transaction naming issues that hide real problems

Hands-on: *Isolate the root cause of a slow checkout transaction using traces and call trees*

Module 2: Distributed Tracing in Microservices

- Why distributed tracing matters for multi-service production incidents
- W3C Trace Context, span propagation, and common broken-trace causes
- Trace visualization: waterfall view, span attributes, and error spans
- Head-based vs tail-based (Infinite Tracing) sampling – what you see and what you miss
- Querying Span data with NRQL to find slow or failing downstream calls

Hands-on: *Follow a failed request end-to-end across services and pinpoint the failing hop*

Module 3: Service Maps, Dependencies, and Correlated Telemetry

- Service maps and cross-application dependencies during incident triage
- Upstream vs downstream impact analysis and blast-radius assessment
- Logs in context: jumping from a trace or error to the related log lines
- Change tracking and deployment markers: correlating releases with regressions

Hands-on: *Correlate a post-deployment latency spike with traces, logs, and the change record*

Day 2: Error Analytics, Alerting, and Incident Response

Module 4: Error Analytics and Errors Inbox

- Capturing errors, exception details, stack traces, and error attributes
- Errors Inbox: grouping, triaging, assigning, and resolving error groups
- Expected vs unexpected errors and reducing error noise
- Error rate trends by host, version, and transaction using NRQL facets

Hands-on: *Triage a burst of exceptions and identify the introducing release*

Module 5: Alert Conditions for Production Support

- Static vs baseline (anomaly) thresholds – choosing the right one per signal
- Signal loss, gap filling, aggregation windows, and evaluation delay
- NRQL alert conditions for business-critical and composite signals
- Tuning noisy alerts: priority, duration, and threshold review techniques

  Hands-on: *Convert a noisy static alert into a tuned baseline NRQL condition*

Module 6: Incident Intelligence and Response Workflow

- Alert policies, issues, incidents, and how they roll up
- Correlation and decisions: reducing duplicate issues during an outage
- Workflows and muting rules for maintenance windows
- Building an incident triage runbook around New Relic signals

  Hands-on: *Run a simulated P1 incident from alert to acknowledgement to root cause*

Day 3: Performance Optimization and Dashboards

Module 7: JVM / Runtime / Infrastructure Metrics

- Thread activity, thread pools, and connection-pool saturation
- Garbage collection time, heap usage, and memory-leak patterns
- CPU, memory, and process-level metrics: correlating host health with APM
- Application health over time: trend analysis and capacity warning signs

  Hands-on: *Diagnose a GC-driven response time degradation using runtime metrics*

Module 8: Database Monitoring and Query Optimization

- DB transactions, database breakdown, and slow query analysis
- Slow query traces, explain plans, and query obfuscation settings
- Database instance views: connections, locks, and throughput
- Common fixes: indexing, N+1 queries, and connection-pool tuning

  Hands-on: *Identify and explain the top slow SQL statement impacting a critical transaction*

Module 9: Production Support Dashboards

- Designing dashboards for on-call: golden signals, service health, and SLA views
- Working with charts, tables, billboards, and thresholds
- Template variables and filters for multi-environment / multi-service views
- NRQL-driven widgets, linked facets, and drill-down navigation
- Sharing, permissions, and dashboard hygiene for support teams

  Hands-on: *Build an on-call command-center dashboard for a microservice application*

Day 4: Synthetic Monitoring – Create, Manage, and Troubleshoot

Module 10: Synthetic Monitor Types and Creation

- Monitor types: ping, simple browser, step monitor, scripted browser, scripted API
- Certificate check and broken links monitors
- Choosing public locations, check frequency, and runtime versions
- Creating monitors and validating results before go-live
- Simulating critical user journeys (login, search, checkout) with scripts

  Hands-on: *Create ping, scripted browser, and scripted API monitors for a sample application*

Module 11: Managing Synthetic Monitors at Scale

- Secure credentials: storing and rotating passwords, tokens, and API keys
- Private locations and job managers for internal / firewalled applications
- Monitor downtimes for planned maintenance and releases
- Runtime upgrades, script maintenance, and bulk management approaches
- SLA reporting and availability tracking with synthetics

Hands-on: *Set up a private location and a monitor using secure credentials for an internal app*

Module 12: Troubleshooting Synthetic Monitors

- Reading check results: waterfall, screenshots, script logs, and error details
- Distinguishing real outages from script, timing, and selector failures
- Flaky monitors: timeouts, dynamic elements, wait strategies, and retries
- Private location issues: job queue, capacity, connectivity, and proxy problems
- Querying SyntheticCheck and SyntheticRequest data with NRQL
- Alerting on synthetics: multi-location conditions to avoid false positives

Hands-on: *Debug a set of intentionally failing monitors and fix their scripts and alerts*

Day 5: Advanced NRQL, Integrations, and Capstone

Module 13: Advanced NRQL for Troubleshooting

- Advanced filtering, WHERE / LIKE / RLIKE, and nested aggregations
- Time comparisons: COMPARE WITH, TIMESERIES, SINCE / UNTIL windows
- percentile(), histogram(), and funnel() for performance and journey analysis
- Data transformation using filter(), FACET CASES, and rate()
- Subqueries and JOINs for cross-dataset investigation

Hands-on: *Write a troubleshooting NRQL query pack for recurring production issues*

Module 14: Integrations and AIOps

- Slack, ServiceNow, and PagerDuty integrations for notification and ticketing
- CI/CD integrations for deployment tracking and change correlation
- Infrastructure integrations: AWS, Azure, and Kubernetes monitoring for support teams
- AI-assisted incident intelligence, anomaly detection, and baselining

Hands-on: *Route a correlated alert to Slack and auto-create a ServiceNow incident*

Module 15: Capstone – Production Support Use Case

- End-to-end troubleshooting of a microservice application under simulated failures
- Build on-call dashboards, tuned alert conditions, and synthetic journey checks
- Define service levels (SLI/SLO), error budgets, and performance reporting strategy
- Present root-cause findings and performance improvement recommendations